REVENUE LEAKAGE INTELLIGENCE SYSTEM

A Multi-Source Analytical Framework for Quantifying, Classifying and Prioritising Revenue Loss

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1. Executive Summary

This report presents a comprehensive analysis of revenue leakage across four distinct operational channels — discount abuse, pricing inconsistency, billing reconciliation failures, and customer return behaviour. Using three real-world datasets and a purpose-built Python-SQLite analytical pipeline, the system quantifies a total identifiable revenue leakage of **$1,613,323** and delivers a prioritised recovery roadmap for leadership action.

The analysis establishes that returns leakage is the dominant loss channel, contributing 64.1% of total identified leakage — yet it remains the channel with the lowest organisational visibility. Pricing leakage, while less dramatic at 27.3%, is highly recoverable through targeted pricing governance. Discount abuse and billing gaps, though smaller in absolute terms, are immediately addressable with standard process controls.

$1.61M	$1.03M	$440K	$138K	$341
Total Revenue Leakage	Returns Leakage	Pricing Leakage	Discount Leakage	Billing Leakage

Key Strategic Takeaway: The business is losing more revenue through uncontrolled return behaviour than through all other leakage channels combined. A structural intervention in return management policy will deliver the highest recovery impact.

2. Project Overview

2.1 Objectives

The Revenue Leakage Intelligence System was designed to address a fundamental gap in organisational financial visibility: the absence of a systematic, data-driven framework for detecting, quantifying, and prioritising revenue loss. The system pursues four core objectives:

- Detect revenue leakage events across multiple transaction types using rule-based analytical logic
- Quantify total leakage in absolute monetary terms, attributable to specific products, customers, and categories
- Classify leakage by business severity to enable resource-efficient remediation prioritisation
- Deliver an executive-ready dashboard that translates analytical findings into actionable business intelligence

2.2 Technical Architecture

Component	Description	Output
Notebook 01	Superstore Discount & Pricing Leakage Analysis	superstore_analysis table
Notebook 02	Olist Billing Reconciliation Analysis	olist_billing_analysis table
Notebook 03	UCI Returns Leakage & Customer Risk	uci_customer_return_analysis table
Notebook 04	Master KPI Consolidation & Export	master_kpi_summary table + 4 CSVs
Power BI	3-Page Executive Dashboard	Executive, Operational, Risk views

2.3 Technology Stack

Layer	Tools Used
Data Processing	Python 3 · Pandas · NumPy
Statistical Logic	NumPy percentile functions · custom rule engines
Database	SQLite (revenue_leakage.db) — shared across all notebooks
Visualisation	Matplotlib · Seaborn (notebook) · Power BI (dashboard)
Data Export	CSV export to Data/Processed_Data/ for Power BI ingestion

3. Datasets

3.1 Sample Superstore (Kaggle)

The Sample Superstore dataset is a widely used retail simulation dataset covering US-based transactions across three major product categories. It serves as the source for both discount leakage and pricing leakage analyses.

Property	Detail
Source	Kaggle — Sample Superstore Dataset
Size	9,994 rows x 21 columns
Geography	United States
Time Period	2014 – 2017
Categories	Furniture, Office Supplies, Technology
Key Columns	Sales, Quantity, Discount, Profit, Product ID, Category, Sub-Category, Region
Leakage Types	Discount Abuse Leakage + Pricing Leakage

3.2 Olist Brazilian E-Commerce (Kaggle)

The Olist dataset is a multi-table relational dataset from a Brazilian e-commerce platform. Three tables — orders, order items, and payments — are merged to construct the billing reconciliation analysis.

Table	Rows (approx.)	Key Columns
olist_orders_dataset	99,441	order_id, order_status, order_delivered_customer_date
olist_order_items_dataset	112,650	order_id, price, freight_value
olist_order_payments_dataset	103,886	order_id, payment_value, payment_type, payment_installments

3.3 UCI Online Retail II

The UCI Online Retail II dataset is a transaction log from a UK-based online wholesale retailer. It uniquely includes return transactions encoded as negative-quantity invoices, making it ideal for returns leakage and customer risk segmentation analysis.

Property	Detail
Source	UCI ML Repository — Online Retail II
Size	~1,067,371 rows x 8 columns
Geography	United Kingdom (primarily)
Time Period	December 2009 – December 2011
Return Signal	Quantity < 0 (verified via 'C'-prefix cancellation invoices)
Key Columns	Invoice, StockCode, Quantity, Price, Customer ID, Country
Leakage Type	Returns Leakage + Customer Return Risk Segmentation

4. Methodology

4.1 Discount Leakage Detection (Notebook 01)

Discount leakage is identified at the transaction level using a dual-condition flag that captures both excessive discounting and confirmed financial loss in the same transaction.

Flag Condition: Discount % > 20 AND Profit < 0

Leakage Amount: abs(Profit) for all flagged transactions

The 20% discount threshold is based on domain convention representing the commonly accepted breakeven boundary for retail margin management. It is acknowledged that this threshold is not statistically derived from the dataset's own margin distribution and should be recalibrated for business-specific contexts.

4.2 Pricing Leakage Detection (Notebook 01)

Pricing leakage captures revenue foregone when identical products are sold below their established fair-market price — an internally consistent benchmark derived from the dataset itself.

Benchmark Price: 90th percentile Unit Price per Product ID

Leakage Formula: (Benchmark Price - Actual Unit Price) x Quantity

Constraint: Clipped at zero (transactions above benchmark are not treated as negative leakage)

The 90th percentile was selected over mean or median to provide a robust, outlier-resistant benchmark that reflects the upper achievable price for each SKU rather than the average realised price. This approach is directionally conservative — it will understate leakage relative to true market pricing.

4.3 Billing Leakage Detection (Notebook 02)

Billing leakage is identified through a three-table join that reconciles expected payment value (derived from order and freight data) against actual payment amounts received.

- Aggregate item-level data: SUM(price) + SUM(freight_value) per order_id
- Aggregate payment data: SUM(payment_value) per order_id
- Join both aggregations to the orders master table via LEFT JOIN on order_id
- Compute: expected_payment = total_order_value + total_freight_value
- Compute: payment_gap = expected_payment - total_payment_value
- Flag: Billing Leakage = 1 where payment_gap > 1 (threshold eliminates floating-point noise)

4.4 Returns Leakage & Customer Risk (Notebook 03)

Returns leakage is quantified using the negative-quantity signal native to the UCI dataset. A customer-level risk model is built on top of aggregated return behaviour.

Return Rate %	Risk Category	Interpretation
<= 5%	Low Risk	Normal return behaviour — no intervention required
5% – 15%	Medium Risk	Elevated returns — monitor and investigate drivers
> 15%	High Risk	Abnormal behaviour — flag for policy review or blacklisting

> 100%	Extreme	Return value exceeds purchase value — potential fraud indicator

4.5 Master KPI Consolidation (Notebook 04)

All upstream leakage amounts are extracted from SQLite and assembled into a single master KPI table with severity classification and recovery timeline mapping.

Contribution %	Severity	Recovery Category	Indicative Timeline
< 10%	Low	Recoverable	0 – 3 Months
10% – 30%	Medium	Process Fix	3 – 6 Months
>= 30%	Critical	Structural	6 – 12 Months

5. Key Findings

5.1 Consolidated Leakage Summary

Leakage Type	Amount ($)	Contribution %	Severity	Recovery
Returns Leakage	10,34,085.87	64.10%	Critical	Structural (6–12 Months)
Pricing Leakage	4,40,380.54	27.30%	Medium	Process Fix (3–6 Months)
Discount Leakage	1,38,515.24	8.59%	Low	Recoverable (0–3 Months)
Billing Leakage	341.03	0.02%	Low	Recoverable (0–3 Months)
TOTAL	16,13,322.68	100.01%	—	—

5.2 Returns Leakage — Critical Priority

Returns leakage is the single largest revenue loss channel, accounting for 64.1% of total identified leakage at $1,034,086. The customer risk segmentation reveals both systematic over-returning and individual cases that exhibit potential fraud behaviour.

High-Risk Customer Summary (Top 10):

Customer ID	Sales Amount ($)	Return Amount ($)	Return Rate %	Risk
14380	48.96	148.69	303.70%	High Risk
14255	1,000.63	2,442.23	244.07%	High Risk
14328	445.05	890.11	200.00%	High Risk
12918	10,953.50	21,907.00	200.00%	High Risk
13290	208.63	417.26	200.00%	High Risk
14213	1,192.20	2,384.40	200.00%	High Risk
14802	1,502.98	3,005.96	200.00%	High Risk
15802	451.42	902.84	200.00%	High Risk
16252	295.09	590.18	200.00%	High Risk
14906	68.44	135.42	197.87%	High Risk

Customer 14380 exhibits a return rate of 303.7% — returning significantly more value than was ever purchased. This pattern, combined with Customer 12918's $21,907 in returns on $10,954 in sales, suggests that at least a subset of high-risk customers may be exploiting return policies rather than returning due to genuine product dissatisfaction.

5.3 Pricing Leakage — Medium Priority

Pricing leakage totals $440,381, representing 27.3% of identified leakage. The analysis reveals that Office Supplies — particularly binding system products — are the dominant pricing leakage contributors, suggesting a category-level pricing governance problem rather than isolated SKU anomalies.

Rank	Product Name	Category	Pricing Leakage ($)
1	GBC DocuBind P400 Electric Binding System	Office Supplies	15,651.39

2	GBC DocuBind TL300 Electric Binding System	Office Supplies	13,365.15
3	GBC Ibimaster 500 Manual ProClick Binding System	Office Supplies	13,241.05
4	Fellowes PB500 Electric Punch Comb Binding Machine	Office Supplies	11,947.31
5	Logitech G19 Programmable Gaming Keyboard	Technology	10,800.95
6	Canon imageCLASS 2200 Advanced Copier	Technology	8,399.98
7	Cubify CubeX 3D Printer Double Head Print	Technology	8,399.97
8	Lexmark MX611dhe Monochrome Laser Printer	Technology	8,261.95
9	Ibico EPK-21 Electric Binding System	Office Supplies	8,089.16
10	Imation 16GB Mini TravelDrive USB Flash Drive	Technology	7,747.51

5.4 Discount Leakage — Low Priority

Discount leakage totals $138,515, representing 8.59% of total identified leakage. The scatter analysis of Discount % vs Profit Margin % confirms a strong negative relationship — as discount levels exceed 40%, profit margins consistently enter deeply negative territory, with extreme cases recorded at -275% margin.

Sub-Category	Discount Abuse Amount ($)	% of Discount Leakage
Binders	39,000+	~28%
Tables	31,000+	~22%
Machines	30,000+	~22%
Bookcases	11,000+	~8%
Appliances	9,000+	~7%
Chairs	7,000+	~5%
Phones	7,000+	~5%

Binders, Tables, and Machines together account for approximately 72% of all discount leakage, indicating that these sub-categories likely have the weakest discount approval controls or the highest competitive price pressure from the sales team.

5.5 Billing Leakage — Low Priority

Billing leakage is the smallest identified channel at $341 total, representing 0.02% of aggregate leakage. While the absolute value is low, the existence of orders with complete payment failures (payment received = $0 despite expected value of $143.46) signals potential payment processing infrastructure gaps that merit automated monitoring.

Order ID (truncated)	Payment Gap ($)	Expected ($)	Received ($)
bfbd0f9bdef84302...	143.46	143.46	0.00
262118ce178bb3e4...	51.62	177.74	126.12
fd33085945f15975...	21.80	234.62	212.82
6e57e23ecac1ae88...	16.50	350.41	333.91
4154bf1348caac78...	14.99	165.26	150.27

6. Power BI Dashboard

The analytical outputs are delivered through a three-page Power BI dashboard designed for distinct decision-maker audiences at each layer of the organisation.

Page	Title	Primary Audience	Key Visuals
1	Executive Overview	C-Suite / Leadership	5 KPI Cards · Leakage Contribution Donut · Severity Bar Chart · Recovery Action Matrix Table
2	Operational Leakage Analysis	Sales & Category Management	Top 10 Pricing Leakage Products Table · Discount-to-Profit Margin Scatter · Discount Leakage by Sub-Category
3	Return & Billing Risk	Finance & Risk Management	Customer Return Risk Donut · High-Risk Customer Table · Payment Reconciliation Gap Table · Billing Distribution Histogram

The dashboard is connected directly to the four CSV exports from Notebook 04. The Page Navigator component allows stakeholders to move between the three views while retaining any active filter context applied at the page level.

7. Assumptions & Limitations

#	Assumption / Limitation	Impact on Analysis
1	20% discount threshold based on domain convention, not statistical derivation	Threshold sensitivity varies by business; may overstate or understate leakage in non-retail contexts
2	90th percentile Unit Price used as fair-value pricing benchmark	Dataset-internal only; does not reflect external market pricing — directionally conservative
3	Payment gap > $1 threshold to exclude processor rounding noise	Genuine micro-gaps below $1 are excluded; immaterial in practice
4	Total of $1.61M aggregates USD + BRL + GBP values	Conceptual framework total only — not a single-entity P&L; figure; do not use for financial reporting
5	Customers with return rate > 100% are included without adjustment	May represent data truncation, fraud, or wholesale aggregation; warrants separate investigation
6	Cancelled-status orders included in billing gap analysis	Some identified gaps may represent legitimate refunds or authorised cancellation adjustments
7	Recovery timelines are heuristic business-convention estimates	Actual timelines will vary by organisational change capacity and data infrastructure maturity
8	Returns are not matched back to original sale invoices	Some return amounts may relate to purchases outside the dataset time window

8. Recovery Action Roadmap

The following roadmap translates analytical findings into prioritised business actions, ordered by recovery timeline and complexity of organisational change required.

Priority	Leakage Type	Amount	Recommended Actions	Timeline
1 (Highest)	Returns Leakage	$1.03M	Implement return rate monitoring per customer. Establish high-risk customer flag and escalation workflow. Introduce return reason validation at point of reversal. Consider return rate caps for repeat high-volume returners.	6–12 Months
2	Pricing Leakage	$440K	Enforce automated pricing floor controls at transaction entry. Establish minimum price policy per SKU based on 90th percentile benchmark. Conduct quarterly pricing audit for Office Supplies category. Train sales team on pricing governance compliance.	3–6 Months
3	Discount Leakage	$138K	Implement two-tier discount approval: auto-approve <= 20%, manager approval for 20–35%, director approval above 35%. Configure real-time margin guardrail alerts. Prioritise Binders, Tables, and Machines sub-categories.	0–3 Months
4	Billing Leakage	$341	Deploy automated payment reconciliation alerts for gaps > $1. Investigate zero-payment orders immediately upon identification. Add payment gap KPI to existing finance monitoring dashboard.	0–3 Months

Total Recoverable Leakage: $1,613,323 — addressable through a structured 12-month remediation programme.

9. Conclusion

The Revenue Leakage Intelligence System demonstrates that systematic revenue loss is neither inevitable nor invisible — it is quantifiable, attributable, and recoverable when the right analytical framework is applied.

The project's primary contribution is not the identification of any single leakage event, but the construction of a replicable, scalable pipeline that any organisation can adapt to its own transaction data. The SQLite-to-Power BI architecture is deliberately lightweight — it can be migrated to enterprise data warehouse infrastructure with minimal structural change.

The findings reinforce a counter-intuitive but important business insight: the largest source of revenue leakage — returns — is the one that receives the least analytical attention in most organisations. A $1M structural problem hiding behind a routine operational process is exactly the kind of insight that data analytics is positioned to surface.

From reactive reporting to proactive revenue protection — that is the value this system delivers.

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Datasets: Sample Superstore (Kaggle) · Olist Brazilian E-Commerce (Kaggle) · UCI Online Retail II